A. Proofs of Convergence of SnareNet

The convergence analysis of SnareNet is based on the observation that SnareNet's repair layer update with $\lambda > 0$ is equivalent to preconditioned gradient descent (PGD) on the constraint violation function defined as

$$F(y) = \frac{1}{2}\|g(y) - \mathcal{P}_{\mathbf{B}(\ell,u)}(g(y))\|^2. \tag{1}$$

Appendix A.1 establishes the analytical properties of F that are crucial for the convergence analysis. The convergence analysis for linear and convex constraints follows in Appendix A.2 and Appendix A.3, respectively.

A.1. Analytical Properties of F

The function F is twice differentiable almost everywhere, despite the presence of the seemingly non-smooth projection operator $\mathcal{P}_{\mathbf{B}(\ell,u)}$. Moreover, when $g(y) = Ay$ is linear with $A \in \mathbb{R}^{m\times n}$, the function F is convex. See Proposition A.1.

Proposition A.1 (Differentiability & Convexity)**.** *Let $g : \mathbb{R}^n \to \mathbb{R}^m$ be a twice differentiable function. Then $F(y)$ defined in (1) is twice differentiable almost everywhere and its gradient is given by*

$$\nabla F(y) = J_g(y)^\top (g(y) - \mathcal{P}_{\mathbf{B}(\ell,u)}(g(y))). \tag{2}$$

In particular, if $g = A \in \mathbb{R}^{m\times n}$ is linear, the constraint violation function $F(y)$ is convex.

Proof. Observe that $F(y) = R(g(y))$ a composition of g and a separable function $R : \mathbb{R}^m \to \mathbb{R}$ defined by

$$R(z) := \frac{1}{2}\|z - \mathcal{P}_{\mathbf{B}(\ell,u)}(z)\|^2 = \frac{1}{2}\sum_{i=1}^{m}(z_i - \mathcal{P}_{[\ell_i,u_i]}(z_i))^2, \tag{3}$$

where $\mathcal{P}_{[\ell_i,u_i]}$ being the one-dimensional projection operator onto the interval $[\ell_i, u_i]$.

- **Almost everywhere twice differentiability.** It suffices to show that $r_{[\ell,u]}(a) := \frac{1}{2}(a - \mathcal{P}_{[\ell,u]}(a))^2$ with $a \in \mathbb{R}$ is twice differentiable almost everywhere since $R(z) = \sum_{i=1}^m r_{[\ell_i,u_i]}(z_i)$ is the sum of such functions and $F(y) = R(g(y))$ is a composition of R and a twice differentiable g. The piecewise definition of $r(a)$ and its derivatives $r'(a)$ and $r''(a)$ are

$$r_{[\ell,u]}(a) = \begin{cases} \frac{1}{2}(a-\ell)^2, & \text{if } a < \ell; \\ 0, & \text{if } \ell \le a \le u; \\ \frac{1}{2}(a-u)^2, & \text{if } a > u, \end{cases} \qquad r'_{[\ell,u]}(a) = \begin{cases} a-\ell, & \text{if } a < \ell; \\ 0, & \text{if } \ell \le a \le u; \\ a-u, & \text{if } a > u, \end{cases}$$

$$r''_{[\ell,u]}(a) = \begin{cases} 0, & \text{if } \ell < a < u; \\ 1, & \text{if } a < \ell \text{ or } a > u. \end{cases}$$

  Note that $r'_{[\ell,u]}(a) = a - \mathcal{P}_{[\ell,u]}(a)$ is continuous everywhere, and the second derivative $r''_{[\ell,u]}(a)$ is well-defined for all $a \in \mathbb{R}$ except at the two points $a = \ell$ and $a = u$. Thus, $r_{[\ell,u]}(a)$ is twice differentiable almost everywhere, and so is $F(y)$.

- **Gradient expression.** Note that $\nabla R(z) = z - \mathcal{P}_{\mathbf{B}(\ell,u)}(z)$ since $r'_{[\ell,u]}(a) = a - \mathcal{P}_{[\ell,u]}(a)$. The gradient of $F(y)$ follows by the chain rule:

$$\nabla F(y) = J_g(y)^\top \nabla R(g(y)) = J_g(y)^\top (g(y) - \mathcal{P}_{\mathbf{B}(\ell,u)}(g(y))).$$

- **Convexity of F when g is linear.** Observe that $R(z) = \sum_{i=1}^m r_{[\ell_i,u_i]}(z_i)$ is a sum of convex functions since each $r_{[\ell_i,u_i]}$ is convex. Thus, R is convex. As $F(y) = R(Ay)$ is a composition of a convex function R and a linear function Ay, F is convex.

□

Using expression (2) of $\nabla F(y)$, SnareNet's repair update (11) with $\lambda > 0$ can be expressed as

$$\begin{aligned} y^{k+1} &= y^k - (J_g(y^k)^\top J_g(y^k) + \lambda I)^{-1} J_g(y^k)^\top (g(y^k) - \mathcal{P}_{\mathbf{B}(\ell,u)}(g(y^k))) \\ &= y^k - (J_g(y^k)^\top J_g(y^k) + \lambda I)^{-1} \nabla F(y^k). \end{aligned} \tag{4}$$

Equation (4) can be interpreted as a PGD step on $F(y)$ with preconditioner $P_k := (J_g(y^k)^\top J_g(y^k) + \lambda I)^{-1}$. To show the convergence under the choice P_k, we first establish an upper bound on $F(y)$ in Lemma A.2 that is analogous to the quadratic upper bound for smooth functions, and then show that F satisfies the Polyak-Lojasiewicz (PL) condition under the preconditioned norm when $J_g(y)$ has full-row rank in Lemma A.3. Finally, we prove the convergence of SnareNet in Theorem A.4 and Theorem A.5 for linear and nonlinear constraints, respectively.

Lemma A.2 (Upper bound on F)**.** *Let $g : \mathbb{R}^n \to \mathbb{R}^m$ be a twice differentiable function with an L_g-Lipschitz continuous Jacobian. Then for any $x, y \in \mathbb{R}^n$, the constraint violation function $F(y)$ defined in* (1) *satisfies the bound*

$$\begin{aligned} F(y) \le F(x) &+ \langle \nabla F(x), y - x \rangle + \frac{1}{2}(y-x)^\top J_g(x)^\top J_g(x)(y-x) \\ &+ \frac{L_g}{2}\left[\|\nabla R(g(x))\| + \|J_g(x)(y-x)\| + \frac{L_g}{4}\|y-x\|^2 \right] \|y-x\|^2. \end{aligned} \tag{5}$$

In particular, if $g(y) = Ay$ is linear with $A \in \mathbb{R}^{m\times n}$, the bound in (5) *reduces to*

$$F(y) \le F(x) + \langle \nabla F(x), y - x \rangle + \frac{1}{2}(y-x)^\top A^\top A (y-x). \tag{6}$$

Proof. We first show that $R(z)$ defined in (3) is 1-smooth, or equivalently, its gradient $\nabla R(z) = z - \mathcal{P}_{\mathbf{B}(\ell,u)}(z)$ is 1-Lipschitz continuous: for any $z, z' \in \mathbb{R}^m$, we have

$$\begin{aligned} \|\nabla R(z) - \nabla R(z')\|^2 &= \|(z - \mathcal{P}_{\mathbf{B}(\ell,u)}(z)) - (z' - \mathcal{P}_{\mathbf{B}(\ell,u)}(z'))\|^2 \\ &= \|z - z'\|^2 - 2\langle \mathcal{P}_{\mathbf{B}(\ell,u)}(z) - \mathcal{P}_{\mathbf{B}(\ell,u)}(z'), z - z' \rangle + \|\mathcal{P}_{\mathbf{B}(\ell,u)}(z) - \mathcal{P}_{\mathbf{B}(\ell,u)}(z')\|^2 \\ &\le \|z - z'\|^2 - \|\mathcal{P}_{\mathbf{B}(\ell,u)}(z) - \mathcal{P}_{\mathbf{B}(\ell,u)}(z')\|^2 \qquad \text{(by firmly non-expansiveness of } \mathcal{P}_{\mathbf{B}(\ell,u)}) \\ &\le \|z - z'\|^2. \end{aligned}$$

Apply the 1-smoothness of $R(z)$ to upper bound $F(y) = R(g(y))$ by

$$\begin{aligned} F(y) = R(g(y)) &\le R(g(x)) + \langle \nabla R(g(x)), g(y) - g(x) \rangle + \frac{1}{2}\|g(y) - g(x)\|^2 \\ &= F(x) + \langle \nabla R(g(x)), g(y) - g(x) \rangle + \frac{1}{2}\|g(y) - g(x)\|^2. \end{aligned} \tag{7}$$

Let $e_x(y) := g(y) - g(x) - J_g(x)(y-x)$ denote the error of linear approximation. Using the fundamental theorem of calculus and the L_g-Lipschitz continuity of J_g, the error can be bounded by

$$\|e_x(y)\| = \left\| \int_0^1 [J_g(x + t(y-x)) - J_g(x)](y-x)\, dt \right\| \le \int_0^1 L_g t \|y-x\|^2\, dt = \frac{L_g}{2}\|y-x\|^2. \tag{8}$$

Cauchy-Schwartz inequality and (8) imply

$$\begin{aligned} \langle \nabla R(g(x)), g(y) - g(x) \rangle &= \langle \nabla R(g(x)), g(y) - g(x) - J_g(x)(y-x) \rangle + \langle \nabla R(g(x)), J_g(x)(y-x) \rangle \\ &\le \|\nabla R(g(x))\| \|g(y) - g(x) - J_g(x)(y-x)\| + \langle J_g(x)^\top \nabla R(g(x)), y - x \rangle \\ &\le \frac{L_g \|\nabla R(g(x))\|}{2}\|y-x\|^2 + \langle \nabla F(x), y - x \rangle. \end{aligned} \tag{9}$$

Equation (8) also implies

$$\begin{aligned} \|g(y) - g(x)\|^2 &= \|e_x(y) + J_g(x)(y-x)\|^2 \\ &= \|J_g(x)(y-x)\|^2 + 2\langle J_g(x)(y-x), e_x(y) \rangle + \|e_x(y)\|^2 \\ &\le \frac{1}{2}(y-x)^\top J_g(x)^\top J_g(x)(y-x) + L_g\|J_g(x)(y-x)\|\|y-x\|^2 + \frac{L_g^2}{4}\|y-x\|^4. \end{aligned} \tag{10}$$

Combine (7), (9), and (10) to conclude (5). When $g(y) = A(y)$ is linear, the Jacobian of g is constant $J_g(y) = A$ for all $y \in \mathbb{R}^n$ and thus $L_g = 0$. The bound reduces to (6). ☐

Lemma A.3 (PL-condition in preconditioned norm)**.** *Suppose $J_g(y)$ has full-row rank with minimum singular value $\sigma_{\min}(J_g(y)) > 0$ and $\mathcal{C} := \{y \in \mathbb{R}^n \mid \ell \le g(y) \le u\}$ is non-empty. The constraint violation function $F(y)$ defined in (1) satisfies the Polyak-Lojasiewicz (PL) condition under the norm $\|\cdot\|_{(J_g(y)^\top J_g(y)+\lambda I)^{-1}}$:*

$$\frac{1}{2}\|\nabla F(y)\|^2_{(J_g(y)^\top J_g(y)+\lambda I)^{-1}} \ge \frac{\sigma^2_{\min}(J_g(y))}{\sigma^2_{\min}(J_g(y))+\lambda}[F(y) - \min_{y\in\mathbb{R}^n} F(y)] = \frac{\sigma^2_{\min}(J_g(y))}{\sigma^2_{\min}(J_g(y))+\lambda}F(y).$$

Proof. Let $d(y) := g(y) - \mathcal{P}_{\mathbf{B}}(g(y))$. Since $\nabla F(y) = J_g(y)^\top d(y)$, the gradient under preconditioned norm is

$$\|\nabla F(y)\|^2_{(J_g(y)^\top J_g(y)+\lambda I)^{-1}} = d(y)^\top J_g(y)[J_g(y)^\top J_g(y) + \lambda I]^{-1} J_g(y)^\top d(y).$$

Let $\sigma_i(J_g(y))$ denote the i-th singular value of $J_g(y)$. Then the singular values of $J_g(y)[J_g(y)^\top J_g(y) + \lambda I]^{-1}J_g(y)^\top$ are $\frac{\sigma_i^2(J_g(y))}{\sigma_i^2(J_g(y))+\lambda}$ for $i = 1,\dots,m$ with minimum $\frac{\sigma^2_{\min}(J_g(y))}{\sigma^2_{\min}(J_g(y))+\lambda}$. Since $F(y) = \frac{1}{2}\|d(y)\|^2$ and non-empty $\mathcal{C}$ guarantees $\min_{y\in\mathbb{R}^n} F(y) = 0$, we conclude

$$\frac{1}{2}\|\nabla F(y)\|^2_{(J_g(y)^\top J_g(y)+\lambda I)^{-1}} \ge \frac{\sigma^2_{\min}(J_g(y))}{2(\sigma^2_{\min}(J_g(y))+\lambda)}\|d(y)\|^2 = \frac{\sigma^2_{\min}(J_g(y))}{\sigma^2_{\min}(J_g(y))+\lambda}F(y).$$

□

A.2. Linear Convergence of SnareNet for Linear Constraints

Theorem A.4 (Linear Convergence of SnareNet for Full Row Rank Linear Constraints)**.** *Let $g(y) = Ay$ and assume $A \in \mathbb{R}^{m\times n}$ has full row rank with minimum singular value $\sigma_{\min} > 0$. Suppose the feasible set $\mathcal{C} := \{y \in \mathbb{R}^n \mid \ell \le Ay \le u\}$ is non-empty. Then SnareNet with $\lambda > 0$ converges linearly to a feasible point at rate*

$$F(y^k) \le \left(1 - \frac{\sigma^2_{\min}}{\sigma^2_{\min}+\lambda}\right)^k F(y^0).$$

Proof. Denote $H_\lambda := A^\top A + \lambda I$. Then $y^{k+1} = y^k - H_\lambda^{-1}\nabla F(y^k)$ and $A^\top A \prec H_\lambda$. Substitute $y = y^{k+1}$ and $x = y^k$ into (6) to obtain

$$\begin{aligned}
F(y^{k+1}) &\le F(y^k) + \langle \nabla F(y^k), y^{k+1} - y^k\rangle + \frac{1}{2}(y^{k+1}-y^k)^\top A^\top A(y^{k+1}-y^k)\\
&\le F(y^k) + \langle \nabla F(y^k), y^{k+1} - y^k\rangle + \frac{1}{2}(y^{k+1}-y^k)^\top H_\lambda(y^{k+1}-y^k)\\
&= F(y^k) - \nabla F(y^k)^\top H_\lambda^{-1}\nabla F(y^k) + \frac{1}{2}\nabla F(y^k)^\top H_\lambda^{-1}\nabla F(y^k)\\
&\le F(y^k) - \frac{1}{2}\|\nabla F(y^k)\|^2_{H_\lambda^{-1}},
\end{aligned} \tag{11}$$

proving the descent property for SnareNet's update. Since A has full row rank, Lemma A.3 guarantees $F(y)$ satisfies PL-condition and thus

$$F(y^{k+1}) \le F(y^k) - \frac{\sigma^2_{\min}}{\sigma^2_{\min}+\lambda}F(y^k) = \left(1 - \frac{\sigma^2_{\min}}{\sigma^2_{\min}+\lambda}\right)F(y^k).$$

□

A.3. Convergence of SnareNet for Convex Constraints

Theorem A.5 (Convergence of SnareNet for Convex Constraints)**.** *Let $\mathcal{C}$ be a non-empty convex set represented by*

$$\mathcal{C} := \{y \in \mathbb{R}^n \mid h(y) \le 0,\ \ell_A \le Ay \le u_A\}, \tag{12}$$

where $A \in \mathbb{R}^{m_A\times n}$, $\ell_A \in \mathbb{R}^{m_A}$, $u_A \in \mathbb{R}^{m_A}$, and $h : \mathbb{R}^n \to \mathbb{R}^{m_h}$ is convex twice differentiable and has L-Lipschitz continuous Jacobian. Let $y^0 \in \mathbb{R}^n$ be the initial point, $\mathcal{D} := \{y \in \mathbb{R}^n \mid F(y) \le F(y^0)\}$ be the sublevel set, $D :=$

$\sup_{y\in\mathcal{D}}\inf_{y^*\in\mathcal{C}}\|y-y^*\|$, *and* $\sigma^2_{\max}(J_h(y))+\sigma^2_{\max}(A)\le\sigma^2$ *for all* $y\in\mathcal{D}$. *For* $\lambda>0$ *satisfying* $\lambda^2-\sqrt{2F(y^0)}L\lambda+\frac{L^2F(y^0)^2}{16}\ge 0$, *SnareNet converges to a feasible point at rate*

$$F(y^k)\le\frac{F(y^0)}{1+\frac{F(y^0)}{2D^2(\sigma^2+\lambda)}k}. \tag{13}$$

Proof. Observe that F is convex under the convex constraints represented in (12) since F takes the form

$$F(y):=\frac{1}{2}\|\max(0,h(y))\|^2+\frac{1}{2}\|Ay-\mathcal{P}_{\mathbf{B}(\ell_A,u_A)}(Ay)\|^2$$

and both $\|\max(0,h(y))\|^2$ and $\|Ay-\mathcal{P}_{\mathbf{B}(\ell_A,u_A)}(Ay)\|^2$ are convex functions of y. Using SnareNet's notation, the constraint function g, the bounds ℓ,u, and the Jacobian J_g are (defined) as

$$g(y):=\begin{bmatrix}h(y)\\ Ay\end{bmatrix},\ \ell:=\begin{bmatrix}-\infty\\ \ell_A\end{bmatrix},\ u:=\begin{bmatrix}0\\ u_A\end{bmatrix},\ J_g(y):=\begin{bmatrix}J_h(y)\\ A\end{bmatrix}.$$

The Jacobian of g is L-Lipschitz continuous since $J_h(y)$ is L-Lipschitz continuous and A is constant.

Now, we show that our choice of λ guarantees $y^k\in\mathcal{D}$ for all $k\ge 0$ and also the descent property $F(y^{k+1})\le F(y^k)$. We prove by induction. Note that $y^0\in\mathcal{D}$ by definition. Suppose $y^k\in\mathcal{D}$ by induction hypothesis. Substitute $y=y^{k+1}$ and $x=y^k$ into (5) to obtain

$$\begin{aligned}F(y^{k+1})\le{}& F(y^k)+\langle\nabla F(y^k),y^{k+1}-y^k\rangle+\tfrac{1}{2}(y^{k+1}-y^k)^\top J_g(y^k)^\top J_g(y^k)(y^{k+1}-y^k)\\ &+\tfrac{L_g}{2}\left[\|\nabla R(g(y^k))\|+\|J_g(y^k)(y^{k+1}-y^k)\|+\tfrac{L_g}{4}\|y^{k+1}-y^k\|^2\right]\|y^{k+1}-y^k\|^2.\end{aligned} \tag{14}$$

Since $y^k\in\mathcal{D}$, we have $\|\nabla R(g(y^k))\|=\|g(y^k)-\mathcal{P}_{\mathbf{B}(\ell,u)}(g(y^k))\|=\sqrt{2F(y^k)}\le\sqrt{2F(y^0)}=:M$ and thus

$$\|J_g(y^k)(y^{k+1}-y^k)\|\le\left\|J_g(y^k)(J_g(y^k)^\top J_g(y^k)+\lambda I)^{-1}J_g(y^k)^\top\right\|\left\|\nabla R(g(y^k))\right\|\le M, \tag{15}$$

$$\|y^{k+1}-y^k\|^2=\left\|(J_g(y^k)^\top J_g(y^k)+\lambda I)^{-1}J_g(y^k)^\top\right\|^2\left\|\nabla R(g(y^k))\right\|^2\le\tfrac{M^2}{4\lambda}. \tag{16}$$

Let $H_k:=J_g(y^k)^\top J_g(y^k)+\lambda I$. Then SnareNet's update reads $y^{k+1}=y^k-H_k^{-1}\nabla F(y^k)$ and (14) becomes

$$\begin{aligned}F(y^{k+1})&\le F(y^k)+\langle\nabla F(y^k),y^{k+1}-y^k\rangle+\tfrac{1}{2}(y^{k+1}-y^k)^\top J_g(y^k)^\top J_g(y^k)(y^{k+1}-y^k)\\ &\quad+\left(L_gM+\tfrac{L_g^2M^2}{32\lambda}\right)\|y^{k+1}-y^k\|^2 &&\text{(by (15) and (16))}\\ &\le F(y^k)+\langle\nabla F(y^k),y^{k+1}-y^k\rangle+\tfrac{1}{2}(y^{k+1}-y^k)^\top H_k(y^{k+1}-y^k) &&\text{(by }\lambda^2-L_gM\lambda-\tfrac{L_g^2M^2}{32}\ge 0)\\ &=F(y^k)-\tfrac{1}{2}\|\nabla F(y^k)\|^2_{H_k^{-1}},\end{aligned} \tag{17}$$

which guarantees $F(y^{k+1})\le F(y^k)$ and thus $y^{k+1}\in\mathcal{D}$. The claim that $y^k\in\mathcal{D}$ for all $k\ge 0$ is proved by induction and the descent property is established.

Next, we prove the sublinear convergence using the convexity of F and the descent property (17). Since F is convex and the feasible set $\mathcal{C}$ is non-empty, we have $\min_{y\in\mathbb{R}^n}F(y)=0$ and

$$F(y^k)=F(y^k)-\min_{y\in\mathbb{R}^n}F(y)\le\|\nabla F(y^k)\|_{H_k^{-1}}\|y^k-\mathcal{P}_{\mathcal{C}}(y^k)\|_{H_k}. \tag{18}$$

To bound $\|y^k-\mathcal{P}_{\mathcal{C}}(y^k)\|_{H_k}$, observe that $\|H_k\|\le\sigma^2_{\max}(J_g(y^k))+\lambda\le\sigma^2_{\max}(J_h(y^k))+\sigma^2_{\max}(A)+\lambda\le\sigma^2+\lambda$ since $y^k\in\mathcal{D}$, and thus

$$\|y^k-\mathcal{P}_{\mathcal{C}}(y^k)\|^2_{H_k}\le\|H_k\|\|y^k-\mathcal{P}_{\mathcal{C}}(y^k)\|^2\le(\sigma^2+\lambda)D^2. \tag{19}$$

Combine (17), (18), and (19) to obtain

$$F(y^{k+1})\le F(y^k)-\frac{F(y^k)^2}{2D^2(\sigma^2+\lambda)}. \tag{20}$$

If $F(y^{k+1}) = 0$, the desired bound follows immediately. Otherwise, (20) implies

$$\frac{1}{F(y^{k+1})} - \frac{1}{F(y^k)} \geq \frac{F(y^k) - F(y^{k+1})}{F(y^k)F(y^{k+1})} \geq \frac{F(y^k)}{2D^2(\sigma^2 + \lambda)F(y^{k+1})} \geq \frac{1}{2D^2(\sigma^2 + \lambda)}.$$

The telescoping sum yields

$$\frac{1}{F(y^{k+1})} \geq \frac{1}{F(y^0)} + \frac{k+1}{2D^2(\sigma^2 + \lambda)},$$

which implies the desired sublinear convergence bound (13). □

B. Scaling of Computational Costs

Table 1 shows the training time, memory usage, and number of repair iterations taken for SnareNet on non-convex programs (NCPs) of varying sizes. The experiment were run on Intel 6426Y CPU and an NVIDIA L40S GPU constrained to 16 GB of GPU memory per run. Note that SnareNet may use less computational resources for larger problems under a fixed number of constraints since the problems can be less constrained and require fewer repair iterations to achieve feasibility. For example, the 500 and 1000 variable problems with 100 constraints are cheaper (both in time and memory) than the 100 variable problem with 100 constraints. Table 1 indicates that SnareNet is more suitable for problems with $m \ll n$.

Table 1. Total training time (sec) / memory usage / max. repair iterations on NCP datasets of varying sizes.

	100 constraints (50 eq. + 50 ineq.)	500 constraints (50 eq. + 450 ineq.)
100 variables	3674.4 / 2.3GB / 32	3496.9 / 4.4GB / 100
500 variables	869.1 / 1.3GB / 3	6493.8 / 11.6GB / 11
1000 variables	1107.3 / 1.3GB / 3	2724.5 / 3.3GB / 3

C. Ablation Study on Warm-up Strategies

This section performs an ablation study on NCPs for models trained with different warm-up strategies. Each model was trained for 1000 epochs total, with the warm-up strategy applied for the first 500 epochs. The feasibility tolerance of SnareNet was set to 1e-8. Table 2 shows the evaluation metrics on the test set. We summarize the obsrvations as follows: 1) The warm-up strategy is required for models to achieve good performance. 2) Adaptive relaxation (AdaRel) reduces both the maximum and geometric mean of the optimality gap compared to other warm up strategies. 3) The improvement of AdaRel over soft epochs is more pronounced as λ decreases, with the largest difference being on the HardNet results, corresponding to $\lambda = 0$.

Table 2. Evaluation on NCPs test instances by models trained with different warm-up strategies. Values are mean $\pm$ std across 5 seeds.

Method	Warm-up	Max Opt. Gap.	GMean Opt. Gap.	Max Ineq. Error	Gmean Ineq. Error	Max Eq. Error	Gmean Eq. Error
HardNet (i.e. $\lambda = 0$)	None	$(1.66 \pm 0.35) \times 10^{1}$	8.61 ± 1.62	$(2.14 \pm 0.39) \times 10^{-13}$	$(2.72 \pm 0.53) \times 10^{-16}$	$(2.53 \pm 0.71) \times 10^{-13}$	$(2.22 \pm 0.57) \times 10^{-14}$
HardNet (i.e. $\lambda = 0$)	Soft Epochs	9.34 ± 3.40	1.73 ± 0.58	$(5.88 \pm 1.42) \times 10^{-14}$	$(1.03 \pm 0.00) \times 10^{-16}$	$(9.55 \pm 1.07) \times 10^{-14}$	$(8.50 \pm 0.39) \times 10^{-15}$
HardNet (i.e. $\lambda = 0$)	AdaRel	9.29 ± 8.41	1.01 ± 0.25	$(5.41 \pm 1.10) \times 10^{-14}$	$(1.04 \pm 0.00) \times 10^{-16}$	$(9.04 \pm 0.99) \times 10^{-14}$	$(8.54 \pm 0.28) \times 10^{-15}$
SnareNet ($\lambda = 0.001$)	None	9.52 ± 3.34	6.15 ± 1.75	$(5.67 \pm 1.08) \times 10^{-9}$	$(2.09 \pm 1.67) \times 10^{-15}$	$(8.23 \pm 1.24) \times 10^{-9}$	$(3.24 \pm 1.00) \times 10^{-10}$
SnareNet ($\lambda = 0.001$)	Soft Epochs	3.25 ± 1.55	$(5.84 \pm 1.93) \times 10^{-1}$	$(2.05 \pm 0.28) \times 10^{-9}$	$(1.22 \pm 0.03) \times 10^{-16}$	$(8.14 \pm 1.14) \times 10^{-9}$	$(3.59 \pm 3.57) \times 10^{-10}$
SnareNet ($\lambda = 0.001$)	AdaRel	2.23 ± 1.03	$(3.57 \pm 1.01) \times 10^{-1}$	$(3.36 \pm 2.20) \times 10^{-9}$	$(1.23 \pm 0.04) \times 10^{-16}$	$(6.70 \pm 2.17) \times 10^{-9}$	$(4.08 \pm 1.15) \times 10^{-10}$
SnareNet ($\lambda = 0.1$)	None	5.63 ± 0.49	4.49 ± 0.53	$(1.04 \pm 0.63) \times 10^{-8}$	$(2.90 \pm 0.80) \times 10^{-15}$	$(2.00 \pm 0.61) \times 10^{-8}$	$(2.26 \pm 1.04) \times 10^{-9}$
SnareNet ($\lambda = 0.1$)	Soft Epochs	$(1.90 \pm 0.27) \times 10^{-1}$	$(3.91 \pm 0.44) \times 10^{-2}$	$(1.46 \pm 0.21) \times 10^{-8}$	$(3.23 \pm 0.16) \times 10^{-16}$	$(2.16 \pm 0.31) \times 10^{-8}$	$(4.10 \pm 0.94) \times 10^{-10}$
SnareNet ($\lambda = 0.1$)	AdaRel	$(2.11 \pm 0.25) \times 10^{-1}$	$(3.79 \pm 0.28) \times 10^{-2}$	$(1.39 \pm 0.37) \times 10^{-8}$	$(3.30 \pm 0.13) \times 10^{-16}$	$(2.08 \pm 0.51) \times 10^{-8}$	$(4.08 \pm 1.26) \times 10^{-10}$

D. Soft Constraint Training Can Be Counterproductive

Soft constraint training is commonly employed as a warm-up strategy for hard-constrained neural networks. However, as shown in Figure 1, this approach can be counterproductive. While HardNet achieves a lower optimality gap after 1000 soft epochs than after 500, this reduction does not persist: once hard constraint training begins, the optimality gap increases sharply, negating any apparent progress. Additional soft constraint epochs constitute an inefficient use of computational resources, as the gains from soft constraint training do not transfer to improved final model performance.

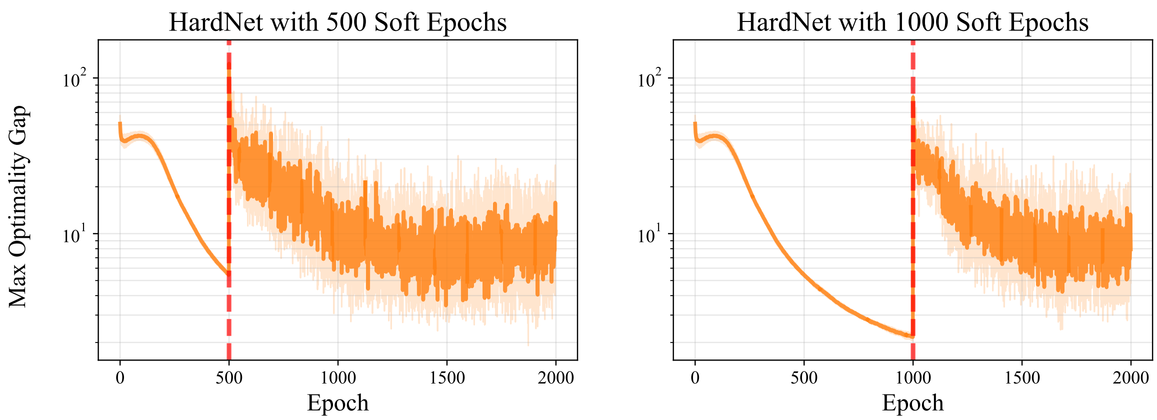
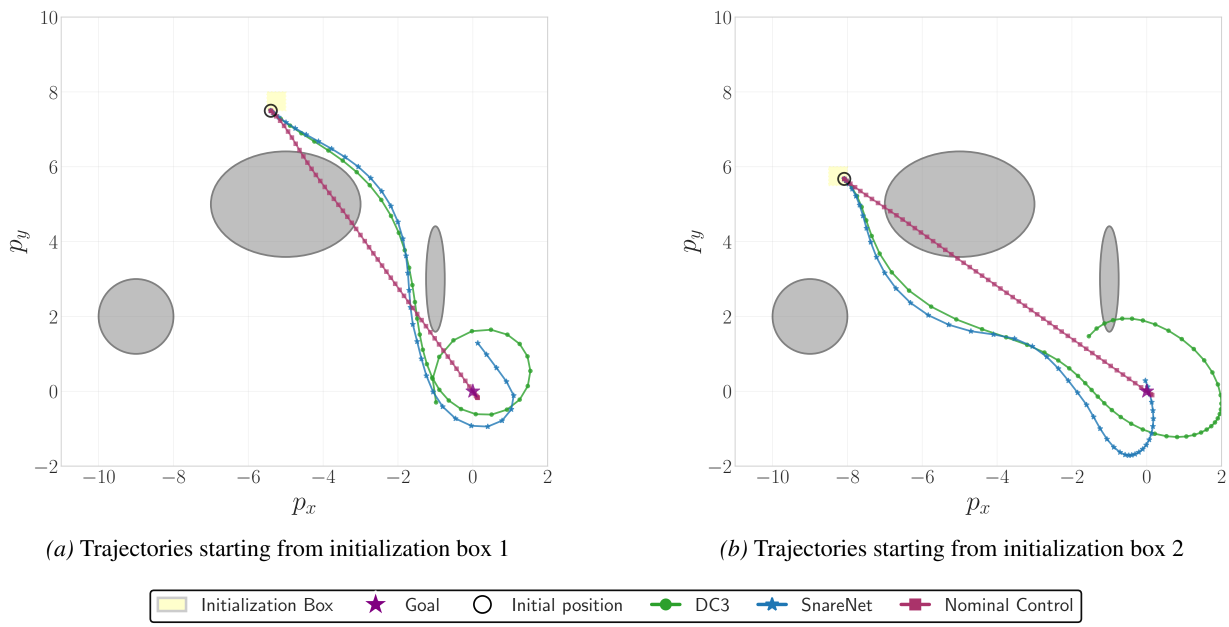


Figure 1. Maximum optimality gap over all NCP validation instances for HardNet trained with soft constraints for 500 and 1000 epochs and hard constraints for the rest.

E. Neural Control Policy Experiment Details

This section includes additional neural control policy experiments with smaller initialization boxes and a longer time horizon. Figure 2 shows the trajectories on datasets with two different initialization boxes, and Table 3 shows the evaluation metrics on the test set. Note that the constraints uses higher order CBFs, so violations of the constraints may not be visually observable as collisions with the obstacles. Even so, DC3's trajectory collides with the obstacle in Figure 2b while SnareNet's trajectory avoids the obstacle in both Figure 2a and Figure 2b. Table 3 justifies better performance of SnareNet compared to DC3, with SnareNet achieving a 99.4% feasibility rate on test instances compared to DC3's 64.9% feasibility rate.



(a) Trajectories starting from initialization box 1

(b) Trajectories starting from initialization box 2

Figure 2. Simulated unicycle trajectories from a test sample inside the initialization region.

Table 3. Evaluation statistics on the CBF test set with 84 samples.

Method	Dataset	Objective	# Feasible (1e-4)	Max Ineq. Error	GMean Ineq. Error
DC3	Box 1	3656	58	6.555	6.785×10^{-15}
SnareNet	Box 1	4060	84	4.21×10^{-8}	1.763×10^{-14}
DC3	Box 2	4269	51	4.17	1.913×10^{-14}
SnareNet	Box 2	5226	83	0.7483	4.038×10^{-14}

F. Experiments on Second-Order Cone Programs (SOCPs)

We added a set of experiments on Second-Order Cone Programs (SOCPs) to demonstrate the generality of SnareNet. SOCPs take the form

$$\begin{array}{ll} \min\limits_{y \in \mathbb{R}^n} & \frac{1}{2} y^T Q y + p^T y \\ \text{s.t.} & \|G_i y + h_i\|_2 \leq C_i^T y + d_i, \quad i = 1, \ldots, m \\ & Ay = x \\ & \ell \leq y \leq u, \end{array}$$

where $x \in \mathbb{R}^{n_{\text{eq}}}$ is the input and the other problem data $(Q, p, G_i, h_i, C_i, d_i, A, \ell, u)$ are constants within the family. Table 4 and Table 5 show the evaluation metrics on the test set for $n = 100$ and $n = 500$ variables, respectively. We observe that SnareNet achieves significantly lower optimality gap than baselines while maintaining the feasibility up to the tolerance level.

Table 4. Evaluation metrics on test set with $n = 100$ variables. Values shown as mean $\pm$ std across 5 random seeds.

Method	Max Opt. Gap	GMean Opt. Gap	GMean Ineq. Error	# Ineq Violations	Max Eq. Error	GMean Eq. Error	# Eq Violations	Test Time (s)
DC3	$(1.71 \pm 1.01) \times 10^1$	$(1.15 \pm 0.55) \times 10^1$	$(1.12 \pm 0.22) \times 10^{-16}$	0.41 ± 0.71	$(1.88 \pm 1.16) \times 10^{-14}$	$(1.48 \pm 0.72) \times 10^{-15}$	0.00	0.043 ± 0.011
HProj	$(1.49 \pm 0.60) \times 10^1$	$(1.14 \pm 0.47) \times 10^1$	$(1.00 \pm 0.00) \times 10^{-16}$	0.00	$(1.84 \pm 1.00) \times 10^{-14}$	$(1.62 \pm 0.79) \times 10^{-15}$	0.00	0.051 ± 0.028
SnareNet ($\texttt{tol} = 10^{-6}$)	$(2.86 \pm 0.24) \times 10^{-2}$	$(8.44 \pm 0.16) \times 10^{-3}$	$(1.96 \pm 0.04) \times 10^{-16}$	0.00	$(7.76 \pm 0.53) \times 10^{-6}$	$(1.13 \pm 0.16) \times 10^{-7}$	0.00	0.557 ± 0.008

Table 5. Evaluation metrics on test set with $n = 500$ variables. Values shown as mean $\pm$ std across 5 random seeds.

Method	Max Opt. Gap	GMean Opt. Gap	GMean Ineq. Error	# Ineq Violations	Max Eq. Error	GMean Eq. Error	# Eq Violations	Test Time (s)
DC3	$(1.60 \pm 0.10) \times 10^2$	$(1.40 \pm 0.03) \times 10^2$	$(1.02 \pm 0.04) \times 10^{-16}$	0.29 ± 0.61	$(7.56 \pm 5.44) \times 10^{-14}$	$(5.83 \pm 4.36) \times 10^{-15}$	0.00	0.194 ± 0.000
HProj	$(1.45 \pm 0.14) \times 10^2$	$(1.35 \pm 0.11) \times 10^2$	$(1.00 \pm 0.00) \times 10^{-16}$	$(2.40 \pm 5.37) \times 10^{-4}$	$(8.64 \pm 5.11) \times 10^{-14}$	$(6.64 \pm 3.11) \times 10^{-15}$	0.00	1.191 ± 0.153
SnareNet ($\texttt{tol} = 10^{-6}$)	$(1.27 \pm 0.54) \times 10^{-2}$	$(9.43 \pm 4.94) \times 10^{-3}$	$(2.73 \pm 0.10) \times 10^{-16}$	0.00	$(3.00 \pm 0.37) \times 10^{-7}$	$(4.48 \pm 0.79) \times 10^{-8}$	0.00	2.179 ± 0.002

G. Illustrate SnareNet on Example 1

This section illustrate the repair update of SnareNet using the concrete example in Example 1. Recall the constraint restricting $y \in \mathbb{R}^2$ to the intersection of two disks of radius 1.5, centered at $(-1, 0)$ and $(1, 0)$:

$$g(y) = \begin{bmatrix} g_1(y) \\ g_2(y) \end{bmatrix} := \begin{bmatrix} (y_1+1)^2 + y_2^2 \\ (y_1-1)^2 + y_2^2 \end{bmatrix} \leq u := \begin{bmatrix} 9/4 \\ 9/4 \end{bmatrix}.$$

The Jacobian of g is given by

$$J_g(y) = \begin{bmatrix} 2(y_1+1) & 2y_2 \\ 2(y_1-1) & 2y_2 \end{bmatrix}.$$

Setup. We initialize the repair process with the infeasible prediction $\hat{y} = (-1, 0) = y^0$, which satisfies the first constraint ($g_1(y^0) = 0 \leq 2.25$) but severely violates the second ($g_2(y^0) = 4 \not\leq 2.25$). We set the Levenberg-Marquardt regularization parameter to $\lambda = 1$ and use no adaptive relaxation to simplify the illustration.

Iteration 0 ($k = 0$). SnareNet projects the evaluation $g(y^0) = (0, 4)$ and calculates the Jacobian at y^0:

$$z^0 = \mathcal{P}_{\mathbf{B}(\ell,u)}(g(y^0)) = (0, 2.25), \quad J_g(y^0) = \begin{bmatrix} 2(-1+1) & 0 \\ 2(-1-1) & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -4 & 0 \end{bmatrix}.$$

The reguarized pseudoinverse operator $J_\lambda^\dagger(y^0) = [J_g(y^0)^\top J_g(y^0) + \lambda I]^{-1} J_g(y^0)^\top$ with $\lambda = 1$ is

$$J_1^\dagger(y^0) = \left(\begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ -4 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{4}{17} \\ 0 & 0 \end{bmatrix}.$$

Then the next iteration y^1 is computed by

$$y^1 = y^0 - J_1^\dagger(y^0)(g(y^0) - z^0) \approx (-0.5882, 0).$$

Iteration Table. Continuing this process demonstrates the convergence to the feasible boundary at $y^* = (-0.5, 0)$. The numerical progression up to $k = 5$ (rounded to four decimal places) is summarized in Table 6.

Table 6. SnareNet repair updates with $\lambda = 1$ on Example 1.

k	y^k	$g(y^k)$	z^k	$\|y^k - y^*\|$
0	$(-1.0000, 0)$	$(0.0000, 4.0000)$	$(0.0000, 2.2500)$	0.5000
1	$(-0.5882, 0)$	$(0.1696, 2.5225)$	$(0.1696, 2.2500)$	0.0882
2	$(-0.5147, 0)$	$(0.2355, 2.2943)$	$(0.2355, 2.2500)$	0.0147
3	$(-0.5026, 0)$	$(0.2474, 2.2578)$	$(0.2474, 2.2500)$	0.0026
4	$(-0.5005, 0)$	$(0.2495, 2.2515)$	$(0.2495, 2.2500)$	0.0005
5	$(-0.5001, 0)$	$(0.2499, 2.2503)$	$(0.2499, 2.2500)$	0.0001